

ECOHARVEST PLATFORM

End-to-End User Acceptance Testing (UAT) Specification & Verification Matrix Customer • Farmer • Admin Modules

Project Name:	EcoHarvest — Decentralized Agritech Marketplace & Ecosystem
Architecture & Tech Stack:	React Native (Expo v57) Web/Mobile, Express.js Backend, MongoDB Atlas, Google Gemini 2.5 AI
Testing Scope & Modules:	Customer Module, Farmer Module, and Admin Portal Module (Fully Synchronized)
Execution Status & Result:	PASS (100% Passed — 60/60 Test Cases Verified Across End-to-End Workflows)

1. Executive Summary & Quality Assurance Overview

This User Acceptance Testing (UAT) document provides comprehensive, rigorous test case specifications and validation results for the EcoHarvest platform. All test scenarios have been systematically executed against the active production-grade React Native / Web client, synchronized Node.js/Express backend, MongoDB Atlas real-time database, and Google Gemini AI services. All 60 test cases across Customer, Farmer, and Admin modules have achieved 100% successful execution (PASS) with zero regressions, validating end-to-end data integrity, real-time escrow synchronization, role-based workflows, and cross-platform reliability.

2. Customer Module — User Acceptance Test Cases

Validates the consumer end-to-end journey including customer onboarding, marketplace search with Gemini AI recommendations, cart/checkout with escrow payment simulation, real-time order tracking with live driver routing, direct farmer messaging, B2B bulk orders, and help desk ticketing.

Test ID	Feature Component	Action / Step-by-Step Instructions	Status
UAT-CUST-001	Customer Registration & Role Provisioning	Navigate to Register Screen -> Select 'Customer' account type -> Enter Name, Email, Phone, and Password -> Tap 'Create Account' -> Verify successful account creation, auth token generation, and immediate redirect to Marketplace.	PASS
UAT-CUST-002	Customer Profile & Multi-Address Setup	Navigate to Profile Screen -> Enter default delivery address, contact coordinates, and preferences -> Tap 'Save Changes' -> Verify persistent storage in MongoDB and automatic address pre-population during checkout.	PASS
UAT-CUST-003	Marketplace Catalog Browsing & Category Filtering	Open Marketplace -> Tap category pills ('Vegetables', 'Fruits', 'Grains', 'Organic') -> Verify catalog instant-filtering with correct product counts, high-resolution media, and farmer tags.	PASS
UAT-CUST-004	Gemini AI Smart Search & Natural Language Query	In Marketplace search bar, enter natural language query (e.g., 'fresh pesticide-free tomatoes in Colombo') -> Verify Gemini AI / smart filter returns semantically relevant produce with accurate price-per-kg tags.	PASS
UAT-CUST-005	Farmer Detail Inspection & Trust Badge Verification	From product card, tap farmer name -> Open Farmer Detail view -> Verify display of KYC Verified Trust Badge, farm bio, ratings/reviews, location distance, and produce inventory.	PASS
UAT-CUST-006	Product Selection & Dynamic	Select a product (e.g., Organic Carrots) -> Adjust quantity	PASS

	Unit Stepper	using '+' / '-' stepper -> Verify real-time subtotal recalculation, available stock limit enforcement, and minimum order validation.	
UAT-CUST-007	Cart Management & Stock Threshold Check	Add multiple items across different categories to Cart -> Open Cart Screen -> Modify quantities and delete an item -> Verify dynamic recalculation of items subtotal, delivery fee, and total amount.	PASS
UAT-CUST-008	Escrow Checkout & Payment Simulation	From Cart, tap 'Proceed to Checkout' -> Select Delivery Address & Payment Method ('Escrow Card / Online Payment') -> Tap 'Place Order' -> Verify simulated payment processing, escrow lock status, and order creation in MongoDB.	PASS
UAT-CUST-009	Order Creation & Real-Time Synchronized Intake	Verify upon order completion that order appears in Customer Orders list as 'Received' and simultaneously broadcasts to the respective Farmer dashboard and Admin Escrow desk.	PASS
UAT-CUST-010	Live Delivery Tracking & Driver Route Simulation	Open active order -> Tap 'Track Delivery' -> Verify interactive map render with origin farm, delivery destination, live driver icon movement, route polyline, and estimated ETA countdown.	PASS
UAT-CUST-011	Direct Driver & Farmer Contact Trigger	Within Delivery Tracking screen, tap 'Call Driver' and 'Message Farmer' action buttons -> Verify system launches native dialer handler and opens integrated direct chat channel.	PASS
UAT-CUST-012	Real-Time In-App Chat with Farmer	Open Chat Screen with a specific farmer -> Send text inquiries regarding harvest freshness -> Verify immediate timestamped message delivery, unread badge counter, and MongoDB chat persistence.	PASS
UAT-CUST-013	B2B Bulk Produce Quote Request Submission	Navigate to Bulk Orders Screen -> Select produce category, enter required quantity (>500kg), target price per kg, and delivery timeframe -> Submit RFQ -> Verify quotation logged in pending state.	PASS
UAT-CUST-014	Bulk Order Counter-Offer Review & Escrow Funding	Receive farmer quote response -> Review revised price-per-kg and delivery schedule -> Tap 'Accept Quote & Fund Escrow' -> Verify bulk contract status shifts to 'Funded / In-Progress'.	PASS
UAT-CUST-015	Customer Help Desk Ticket Creation	Navigate to Profile -> Support & Help Desk -> Select Ticket Category ('Delivery Delay' / 'Quality Dispute'), enter subject and description -> Tap 'Submit Ticket' -> Verify ticket ID generated and visible in Admin Help Desk.	PASS
UAT-CUST-016	Support Ticket Live Resolution & Updates	Open submitted support ticket -> Verify real-time display of ticket status ('Open' -> 'In Progress' -> 'Resolved') and synchronized display of Admin agent responses.	PASS
UAT-CUST-017	Order Delivery Confirmation & Farmer Payout Trigger	Upon driver handover, tap 'Confirm Delivery Received' on active order -> Verify order updates to 'Completed', and trigger automated escrow release signal to farmer wallet.	PASS
UAT-CUST-018	Customer Feedback & Farmer Rating Submission	After order completion, tap 'Rate & Review' -> Select 5-star rating and submit qualitative feedback -> Verify rating updates farmer overall score and appears on farmer profile.	PASS
UAT-CUST-019	Order Cancellation Workflow (Pre-Dispatch)	Open newly placed order in 'Received' status -> Tap 'Cancel Order' -> Confirm prompt -> Verify order status updates to 'Cancelled', stock is restored in inventory, and simulated escrow refund is issued.	PASS
UAT-CUST-020	Offline Resilience & Local Storage Sync	Trigger network disconnection while browsing -> Add items to cart -> Reconnect network -> Verify cart contents and user session persist without state loss via AsyncStorage / local persistence.	PASS

3. Farmer Module — User Acceptance Test Cases

Validates the farmer-centric workflows including KYC onboarding, farm registration, document submission, Gemini AI crop forecasting, inventory lifecycle management, live order pipeline fulfillment, direct messaging, bulk quote negotiation, and earnings reconciliation.

Test ID	Feature Component	Action / Step-by-Step Instructions	Status
UAT-FARM-001	Farmer Registration & Onboarding Initiation	Register new account with 'Farmer' role -> Land on multi-step Farmer Onboarding Wizard -> Step 1: Input Farm Name, Owner Full Name, Mobile Number, and Province/District -> Tap 'Next'.	PASS
UAT-FARM-002	Farm Location Geotagging & Crop Specialization	Step 2: Enter GPS farm coordinates/address, farm size in acres, and select primary crop types (Vegetables, Fruits, Paddy, Spices) -> Tap 'Next' -> Verify step validation.	PASS
UAT-FARM-003	KYC Verification Document Upload	Step 3: Upload National Identity Card (NIC/Passport) and optional GAP/Organic Certificate images -> Verify client-side media validation, thumbnail rendering, and preview capabilities.	PASS
UAT-FARM-004	Onboarding Submission & Verification Desk Routing	Step 4: Review summary details -> Tap 'Submit for Verification' -> Verify farm profile saved in database with 'Pending' status and instantly queued in the Admin Verification Desk.	PASS
UAT-FARM-005	Farmer Dashboard Analytics & Metric Widgets	Log in as verified farmer -> Open Farmer Dashboard -> Verify display of Gross Sales, Total Orders, Active Listings, Customer Rating average, and monthly revenue graph.	PASS
UAT-FARM-006	Gemini AI Harvest Yield & Market Price Forecast	On Farmer Dashboard, navigate to 'Gemini AI Insights' card -> Select crop type (e.g., Green Chili) -> Tap 'Generate Forecast' -> Verify AI generates price trend predictions, optimal harvest window, and regional demand advisory.	PASS
UAT-FARM-007	Weather Impact & Pest Advisory Integration	On AI Insights tab, inspect real-time localized weather telemetry and AI-generated precautionary advice (e.g., rainfall forecast mitigation, organic pest control).	PASS
UAT-FARM-008	New Product Listing Creation	Navigate to 'Add Product' Screen -> Input Title, Category, Price/kg, Available Stock (kg), Harvest Date, Farm Location, and Organic tag -> Upload produce image -> Tap 'Publish Product'.	PASS
UAT-FARM-009	Product Catalog Synchronization & Instant Live State	Verify published product immediately appears in 'My Products' screen and is discoverable in the Customer Marketplace with correct farmer credentials and pricing.	PASS
UAT-FARM-010	Inventory Quantity Update & Out-of-Stock Alert	In 'My Products', select an item -> Edit available stock to '0' -> Save -> Verify product displays 'Out of Stock' badge and disables 'Add to Cart' button in Customer Marketplace.	PASS
UAT-FARM-011	Product Deletion & Catalog Removal	In 'My Products', swipe/tap 'Delete' on an item -> Confirm deletion dialog -> Verify product is permanently removed from catalog and unlinked from active consumer carts.	PASS
UAT-FARM-012	Incoming Order Notification & Pipeline Intake	Simulate customer placing order for farmer's produce -> Open Farmer Orders Screen -> Verify new order immediately appears in 'Received' tab with customer details and item manifest.	PASS
UAT-FARM-013	Order Fulfillment Pipeline: 'Processing' Transition	Select order in 'Received' status -> Tap 'Accept & Start Processing' -> Verify order status updates to 'Processing' in	PASS

		farmer portal, customer app, and backend database.	
UAT-FARM-014	Order Fulfillment Pipeline: 'Packed' & 'Dispatched'	Advance order status from 'Processing' -> 'Packed' -> 'Dispatched' (assign simulated driver) -> Verify customer app reflects status updates in real-time and enables live delivery tracking.	PASS
UAT-FARM-015	Farmer Direct Chat Communication with Buyer	From order detail card, tap 'Chat with Customer' -> Open chat window -> Reply to customer questions -> Verify bidirectional message delivery and persistent chat history.	PASS
UAT-FARM-016	Farmer B2B Bulk Order RFQ Review	Navigate to 'Bulk Orders' tab -> Inspect incoming B2B buyer request (e.g., 1000kg Pumpkin) -> View buyer target price and requested delivery date.	PASS
UAT-FARM-017	Farmer Bulk Quote Negotiation & Counter-Offer	On buyer bulk RFQ, enter proposed counter price per kg and confirmed harvest date -> Tap 'Send Counter Offer' -> Verify update sent to buyer with negotiation audit trail.	PASS
UAT-FARM-018	Farmer Wallet Balance & Escrow Payout Ingestion	Upon customer delivery confirmation, inspect Farmer Dashboard Earnings widget -> Verify escrow funds unlocked and credited to Farmer Net Balance after platform commission deduction.	PASS
UAT-FARM-019	Farmer Profile Customization & Farm Bio Update	Navigate to Profile Screen -> Update Farm description, operating hours, and profile banner -> Save -> Verify updated farm details visible across public marketplace listings.	PASS
UAT-FARM-020	Farmer Support Ticket Filing	From Farmer Profile -> Tap 'Help & Support' -> Submit ticket regarding logistics delay -> Verify ticket appears with 'Farmer' role tag in Admin Help Desk queue.	PASS

4. Admin Portal Module — User Acceptance Test Cases

Validates the administrative ecosystem controls including farmer verification desk, KYC document audit, trust badge issuance, escrow financial settlements, dispute mediation, centralized support ticketing, and platform-wide ecosystem analytics.

Test ID	Feature Component	Action / Step-by-Step Instructions	Status
UAT-ADM-001	Admin Portal Secure Authentication & Role Guard	Navigate to Admin Portal route (/admin) -> Authenticate with Admin credentials -> Verify authorization guard restricts unauthorized roles and lands on Admin Portal Dashboard.	PASS
UAT-ADM-002	Multi-Tab Navigation & Module Switcher	In Admin Portal, switch between 'Verification Desk', 'Escrow Logistics', 'Help Desk', and 'Ecosystem Analytics' tabs -> Verify smooth tab transitions and independent state preservation.	PASS
UAT-ADM-003	Verification Desk: Farmer KYC Submission Queue	Open 'Verification Desk' -> Verify list of all pending farmer onboarding submissions with Farmer Name, Farm Title, District, Submission Timestamp, and Status Badge.	PASS
UAT-ADM-004	Verification Desk: Document Modal Inspection	Click 'Review Application' on pending farmer -> Open inspection modal -> View full-size National ID (NIC), Farm Certificates, and crop details with zoom/rotation support.	PASS
UAT-ADM-005	Farmer Approval & Instant Trust Badge Issuance	In Review modal, click 'Approve Farmer' -> Enter optional approval remark -> Confirm -> Verify farmer status instantly updates to 'Verified', 'Verified' badge issued, and farmer notified.	PASS
UAT-ADM-006	Farmer Rejection Workflow with Audit Rationale	In Review modal for invalid application, click 'Reject' -> Input mandatory rejection reason (e.g., 'Unclear NIC image') -> Confirm -> Verify status updates to 'Rejected' with reason	PASS

		recorded in DB.	
UAT-ADM-007	Escrow Logistics: Real-Time Vault Ledger Ingestion	Open 'Escrow Logistics' tab -> Verify overview metrics (Total Escrow Vault Held, Released Payouts, Pending Settlements, Dispute Reserves) matching MongoDB transaction records.	PASS
UAT-ADM-008	Escrow Order Transaction Pipeline Audit	Inspect escrow ledger table -> Verify listing of all customer orders with Buyer ID, Farmer ID, Total Order Amount, Escrow Lock Timestamp, and Release State.	PASS
UAT-ADM-009	Manual Escrow Release Authorization	Select completed order with pending payout -> Click 'Authorize Payout Release' -> Confirm override -> Verify escrow funds immediately marked 'Released' and credited to farmer balance.	PASS
UAT-ADM-010	Escrow Dispute Mediation & Customer Refund	Select disputed order -> Review dispute details -> Click 'Approve Refund to Customer' -> Verify escrow funds reverted to customer and order status updated to 'Dispute Resolved - Refunded'.	PASS
UAT-ADM-011	Centralized Help Desk Ticket Queue Ingestion	Open 'Help Desk' tab -> Verify real-time display of all user support tickets submitted across Customer and Farmer modules, sorted by priority and recency.	PASS
UAT-ADM-012	Help Desk Ticket Triage & Priority Filtering	Filter tickets by status ('Open', 'In Progress', 'Resolved') and category ('Delivery', 'Payment', 'Quality', 'Account') -> Verify instant UI filtering and ticket count badges.	PASS
UAT-ADM-013	Admin Support Ticket Response & Status Update	Select open ticket -> View message thread -> Type official support reply in admin composer -> Set status to 'Resolved' -> Click 'Send & Close' -> Verify update synced to user app.	PASS
UAT-ADM-014	Ecosystem Analytics: High-Level Platform KPIs	Open 'Ecosystem Analytics' tab -> Verify display of Gross Merchandise Volume (GMV), Total Registered Users, Active Farmers, Total Completed Orders, and Platform Revenue.	PASS
UAT-ADM-015	Ecosystem Analytics: Regional Trade & Crop Distribution	Inspect regional analytics chart -> Verify breakdown of top performing agricultural districts (e.g., Nuwara Eliya, Jaffna, Anuradhapura) and primary crop volume distributions.	PASS
UAT-ADM-016	Gemini AI Ecosystem Health & Anomaly Evaluation	On Ecosystem Analytics, trigger 'Gemini AI Health Assessment' -> Verify AI generates automated macro insights on platform pricing stability, supply-demand balance, and fulfillment health.	PASS
UAT-ADM-017	User Management & Role Audit	Inspect system users list -> Verify distinct identification of Customer, Farmer, and Admin accounts with associated registration dates, verification status, and activity logs.	PASS
UAT-ADM-018	System-Wide Product Catalog Moderation	Inspect active marketplace listings from Admin view -> Verify ability to audit, flag, or remove non-compliant agricultural listings in violation of platform standards.	PASS
UAT-ADM-019	Real-Time WebSocket / DB Synchronization Integrity	Perform simultaneous actions in Customer/Farmer app while Admin portal is open -> Verify Admin UI updates dynamically without requiring manual page reload.	PASS
UAT-ADM-020	Admin Audit Trail & System Log Export	Inspect Admin operations log -> Verify all verification approvals, escrow payouts, and ticket resolutions are recorded with Admin User ID, Action Type, and UTC Timestamp.	PASS

5. Test Execution Summary & Quality Sign-Off

Module Name	Total Test	Passed	Failed / Blocked	Pass Rate (%)
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	Cases			
Customer Module	20	20	0	100.0%
Farmer Module	20	20	0	100.0%
Admin Portal Module	20	20	0	100.0%
Total Ecosystem	60	60	0	100.0%